

Exploring the Impact of Chatbot Design on Student Engagement and Cognitive Load: A Case Study on Error-Analysis Chatbots

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Abstract

While chatbots have been successfully applied in many fields, their potential in education, especially for identifying specific student errors, remains largely underexplored. This study aims to help designers learn more about students' interactions to error analysis chatbots. We evaluated efficiency and reliability of two designs of chatbot step-by-step (i.e., *breaks down complex problems into smaller, manageable questions to mimic the student's thought process*) and Peer-Solution-Check (i.e., *presents fully solved problems for students to assess as either correct or incorrect*). We also measured students' perceived engagement and cognitive load across two kinds of designs. The results showed that the Peer-Solution-Check design consistently outperformed its counterpart in efficiency, engagement, and cognitive load, while maintaining equivalent reliability in error detection. We discussed this study by outlining the challenges encountered and the decisions made, offering insights that can guide HCI practitioners in optimizing chatbot interactions and assessing their effectiveness in real-world educational settings.

CCS Concepts

• Human-centered computing → User centered design.

Keywords

Human-AI interaction, Error analysis, Complex problems

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1 Introduction

Identifying students' error patterns and providing tailored feedback are critical for effective instruction [14]. When educators simply reteach the entire problem-solving process without directly addressing students' misconceptions, those misconceptions can persist, hindering students' progress in more advanced topics [2]. However, manually analyzing errors to uncover patterns, especially for complex tasks, can be labor-intensive and challenging [1]. With recent advancements in generative artificial intelligence [4], chatbots have emerged as tools that can assist in facilitating interactions between humans and machines [3]. Although chatbots have been successfully applied in fields like finance and healthcare [9], their potential in education, especially for identifying specific student errors, remains largely underexplored. To address this gap, we developed an error-analysis chatbot, an AI-driven tool designed to identify and analyze students' error patterns while solving complex problems. This chatbot leverages prior research into error analysis and misconceptions to detect potential mistakes. Two distinct chatbot designs were implemented: the Guided-Problem-Solving design, which guides students through tasks incrementally, and the Peer-Solution-Check design, which presents solved problems for students to judge as correct or incorrect.

In this study, we compare the performance of these two chatbot designs across key performance metrics, including efficiency, reliability, cognitive load, and engagement. Specifically, we evaluate the chatbot's error detection capabilities against those of a human rater and examine how the two designs influence students' perceptions of cognitive load and engagement during their interactions. By streamlining error detection, these AI-driven tools could reduce classroom teachers' workloads and improve learning outcomes. This research provides valuable insights for Human-Computer Interaction (HCI) practitioners seeking to refine educational tools, advancing individualized instruction and promoting educational equity.

2 Background

Error patterns refer to the consistent mistakes students make due to their misconceptions [5]. Identifying specific errors made by students during complex problem-solving, such as mathematical word problems, is particularly challenging because these tasks involve multiple steps and various types of knowledge [8]. Reviewing students' final written answers often does not reveal the specific steps

where they encountered difficulties. To better understand their thinking and pinpoint lingering misconceptions, educators need to observe students closely or conduct strategic interviews [15]. However, students may struggle to clearly explain their problem-solving methods or may be reluctant to share their thought processes. Moreover, classroom teachers, already burdened with numerous responsibilities, often lack the time or specialized training required to conduct such detailed observations and interviews [19].

2.1 Theory-Informed Design

The development of the two distinct error-analysis chatbots designs was informed by established learning theories. The Guided-Problem-Solving approach is rooted in scaffolding theories, which emphasize breaking down complex tasks into smaller, more manageable steps to support learning [20]. By posing sequential and targeted questions, this design encourages active problem-solving and engagement, mirroring traditional teaching practices. Students are guided through the problem-solving process step-by-step, promoting a deeper understanding of each stage [7]. However, this approach places a greater demand on students to generate responses in real time, potentially increasing cognitive load. This elevated cognitive demand can, in some cases, lead to anxiety or fear of making mistakes, particularly when students perceive the questions as high-stakes or overly rigid. While the method effectively fosters active learning, the added pressure it creates may inhibit performance, especially in situations where students feel under stress.

In contrast, the Peer-Solution-Check approach is grounded in cognitive load theory, which seeks to minimize extraneous cognitive demands, enabling learners to focus on understanding problem-solving strategies [21]. This design provides students with fully worked solutions—some correct and some erroneous—and asks them to analyze these examples. By evaluating the solutions and identifying errors, students engage in reflective learning, allowing them to better understand common mistakes and correct problem-solving methods. This approach reduces the cognitive effort required for real-time problem-solving, enabling students to concentrate on diagnosing errors and refining their reasoning [18]. Additionally, by analyzing their selections, the chatbot can effectively diagnose and categorize specific misconceptions, providing valuable insights into students' thought processes.

2.2 Evaluating Chatbot Performance: Efficiency and Reliability

Currently, there is limited understanding of how to design chatbots that enhance student engagement and reduce cognitive load. Yet, student attention and focus play a key role in distinguishing between random errors—caused by distractions or fatigue—and systematic error patterns, which indicate deeper conceptual misunderstandings that require targeted instruction. Factors like cognitive overload or fatigue can impair concentration, leading to random mistakes that don't necessarily reflect genuine misconceptions [10]. If students are not fully engaged, their errors may be the result of inattention rather than actual misunderstandings [10].

Therefore, by ensuring that the chatbot design effectively promotes student engagement and reduces cognitive load, we can be

more confident that the errors identified by the chatbot are meaningful indicators of underlying misconceptions rather than random mistakes.

2.3 Research Questions

In this study, we address two research questions: 1. How does the chatbot's design (Guided-Problem-Solving vs. Peer-Solution-Check) impact its performance in terms of efficiency and reliability when identifying and analyzing students' error patterns? 2. How does the chatbot's design influence students' perceived engagement and cognitive load during their interactions with the chatbot?

2.4 Design of the error-analysis chatbot

Our chatbot design is grounded in prior research on the four critical stages of word-problem solving: general comprehension, understanding mathematical relationships, equation construction, and computations [8]. Each stage represents a key component of problem-solving, and errors at any stage can hinder a student's ability to solve the problem effectively. To address this, the questions in both the Guided-Problem-Solving and Worked-Example designs were developed to diagnose misconceptions corresponding to these stages, as shown in 1.

2.4.1 Guided-Problem-Solving Design. In the Guided-Problem-Solving design, word problems are divided into four stages, with each stage containing a targeted question designed to capture students' misconceptions and error patterns during the problem-solving process:

Question 1 (General Comprehension): This question evaluates their ability to identify the relevant quantities or data required for solving the problem, capturing errors related to general comprehension.

Question 2 (Mathematical Relationships Comprehension): Students are required to identify the phrase that explains the mathematical operation needed to solve the problem, targeting errors in understanding the mathematical relationship.

Question 3 (Equation Construction): Students are asked to construct an equation that expresses the mathematical relationship. This step evaluates their ability to accurately translate the problem into a mathematical equation, capturing errors in equation construction.

Question 4 (Computations): Students are provided with a correct equation and asked to calculate the result.

2.4.2 Peer-Solution-Check Design. In the Worked-Example design, students are presented with correct and incorrect examples of solutions from hypothetical students. Each example highlights a common error associated with a specific stage of the problem-solving process:

Xiao Ming's Solution (General Comprehension): John incorrectly adds unrelated values, demonstrating a lack of understanding of the problem context.

Yue Yue's Solution (Mathematical Relationships): Emma performs subtraction instead of addition, which indicates a misconception about the mathematical relationship between quantities.

Xiao Hai's Solution (Equation Construction): David multiplies the numbers instead of adding them, demonstrating an error in equation construction.

Xi Xi's Solution (Computations): Sophia correctly solves the problem, demonstrating correct computation

3 Methods

3.1 Participants and Procedure

The study was conducted with 50 fifth-grade students from a public school in Western China, consisting of 22 girls and 28 boys with an average age of 11 years ($SD = 0.21$). Prior to participation, both parental or legal guardian consent and child assent were obtained following ethical guidelines. The study protocol was reviewed and approved by the Institutional Review Board (IRB) at the University of Macau, ensuring compliance with ethical standards for research involving minors.

We have communicated the study's purpose, procedures, and potential risks or benefits through both the child assent form and the parent consent form. The children's assent form is written in simple, child-friendly language. It explains the study's goals and the activities involved, such as using the AI chatbot to solve math problems. The form also clearly informs children that participation is voluntary, and they can choose not to participate or withdraw at any time without any consequences or impact on their relationship with the researchers. Additionally, it specifies that if they decide to withdraw, no further data will be collected or used.

One week before the experiment, students completed a paper-and-pencil test containing the same set of word problems used in the study. They were instructed to solve five problems without any time limits. This paper-and-pencil test were rated by human to establish a baseline for comparison. The experiment took place in the school's computer lab during a whole-class session. Students interacted with two different error-analysis chatbots. To ensure fairness and eliminate potential order effects, the sequence in which students interacted with the two chatbot designs was randomized. Half of the students (25) started with the Guided-Problem-Solving design, while the other half (25) began with the Peer-Solution-Check design [11].

Regarding data protection, we have implemented strict protocols to ensure confidentiality. All students' responses and personal data are anonymized, with each child assigned a unique identifier to protect their identity. Additionally, all data, including recordings and responses, are securely stored on password-protected computers, accessible only by the research team. The data will be stored for a maximum of five years, after which it will be completely deleted.

Before using each chatbot, the teacher provided a brief 2-minute introduction to explain the interface and how students should respond. For the Guided-Problem-Solving chatbot, the teacher explained, "The chatbot will ask a series of questions step by step, and you need to type in your answers." For the Peer-Solution-Check chatbot, the teacher clarified, "The chatbot will show you examples of solved problems, and you will need to identify whether each solution is correct or incorrect."

During the lab session, each student was tasked with solving five same mathematics word problems (See Appendix A). Students were given no time limits for completing the problems, ensuring they could work at their own pace. This setup allowed for the accurate collection of performance data, including error patterns, task completion times, and student engagement.

3.2 Training human raters and establishing a reliable baseline for comparison

The two Research assistants (RAs), both master's students specializing in mathematics education, underwent a rigorous training process to ensure consistent and accurate classification of errors. The training began with an initial orientation, during which the RAs were introduced to the study's objectives and provided with an overview of the error-analysis framework. Following the orientation, the RAs participated in guided practice using a curated dataset of student responses. This dataset included pre-classified examples of error patterns, allowing the RAs to practice identifying and categorizing errors under supervision.

After completing the initial training, the RAs independently analyzed all student errors. Their classifications demonstrated a high level of reliability, with an agreement rate of 0.98. Any discrepancies between the two RAs' classifications were identified and documented for further review. The RAs then collaboratively revisited these discrepancies, referencing the study's error classification guidelines and carefully examining the corresponding student responses.

In cases where the RAs could not reach an agreement, a third expert—a faculty member specializing in mathematics education—was consulted to provide an impartial evaluation and final decision. This expert mediation ensured that the resolution process remained systematic and unbiased. The finalized consensus files served as the established baseline for comparing the chatbots' error-detection performance.

3.3 Measures of engagement and cognitive load

The study utilized self-report surveys to measure engagement and cognitive load, with validated scales adapted from previous research. Please refer Appendix B for details.

Engagement was assessed using a survey adapted from Pierce et al. [12], designed to capture both affective and behavioral aspects of student engagement. Affective engagement items focused on students' emotional responses to the chatbot (e.g., "I enjoy talking with Miss Wang on Chatbot A"), while behavioral engagement items measured the degree of attentiveness and effort students invested (e.g., "I carefully read what Miss Wang says on Chatbot A"). Each chatbot design was evaluated with a 10-item scale rated on a 7-point Likert scale. The average Cronbach's alpha for engagement was 0.87, indicating high reliability.

Cognitive load was measured using a survey adapted from Klepsh et al. [6], consisting of 5 items per chatbot design. The scale assessed two dimensions of cognitive load: intrinsic load, which reflects the inherent complexity of the task (e.g., "I do not have the patience to read what Miss Wang says on Chatbot A"), and extraneous load, which captures unnecessary mental effort caused by the interface or instructions (e.g., "I need to read Miss Wang's messages on Chatbot A multiple times before I can respond"). Responses were rated on a 7-point Likert scale, with an average Cronbach's alpha of 0.76.

3.4 Data Analysis

To evaluate the performance of the two chatbot designs, we conducted paired-sample *t*-tests to compare students' task completion

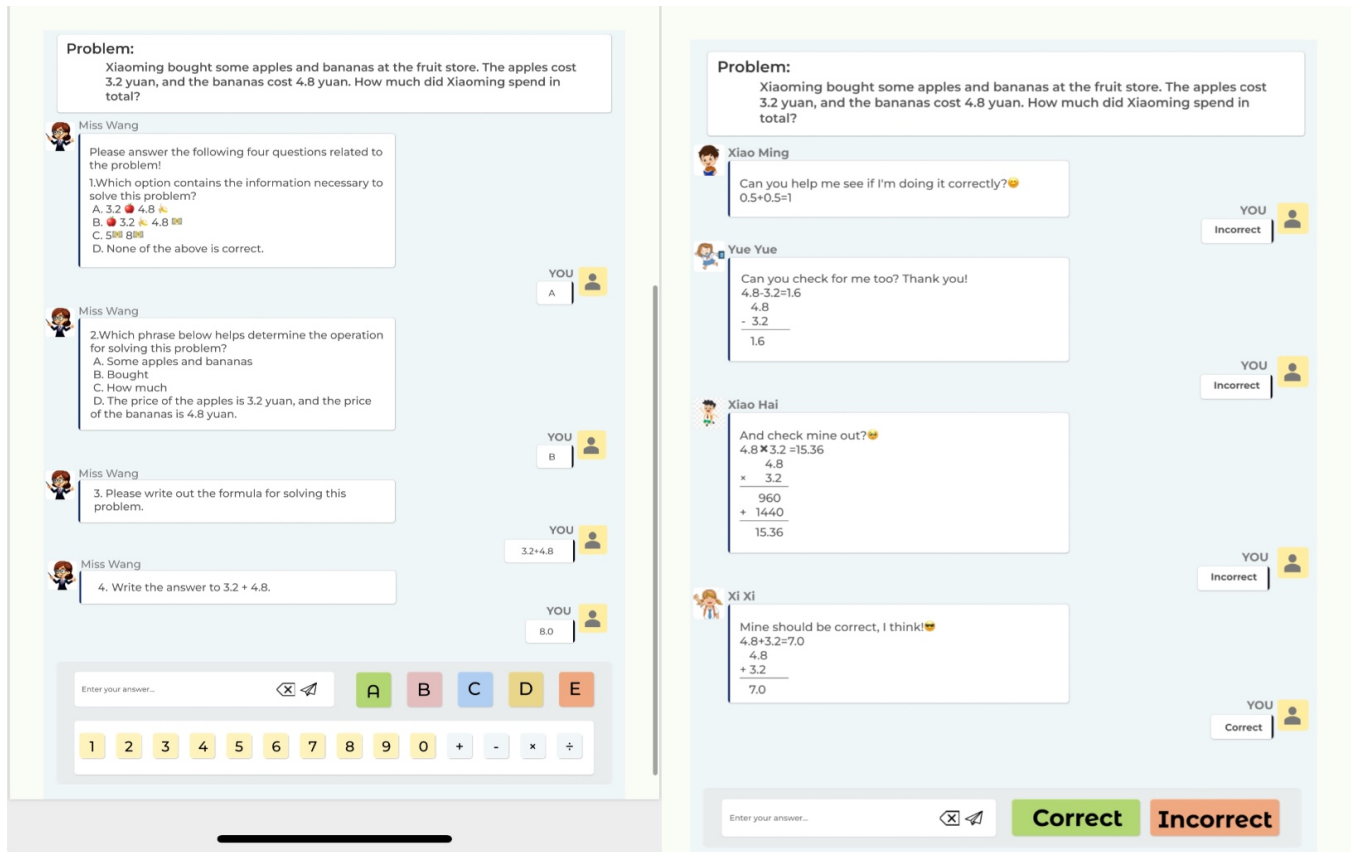


Figure 1: The left shows the interface of Guided-Problem-Solving design and the right shows the interface of Peer-Solution-Check design

times, perceived engagement, and cognitive load between the two designs. This statistical method was chosen because each student experienced both chatbot designs, allowing for direct within-subject comparisons. By treating each student as their own control, the paired-sample t -test minimized between-subject variability, enhancing the precision and reliability of the results. The assumption of normality was checked using the Shapiro-Wilk test, and the results showed that the data for all measures were normally distributed ($p > .05$). Additionally, the assumption of equal variances was verified using Levene's test, which indicated no significant violations ($p > .05$). This confirmed the appropriateness of the paired-sample t -test. The t -test results showed that there were statistically significant differences in task completion times ($p < .01$), engagement levels ($p < .05$), and cognitive load ($p < .01$) between the two chatbot designs, suggesting a clear distinction in their relative effectiveness.

To assess the reliability of the chatbot designs, we compared the error patterns identified by the two chatbot designs with those identified by two RAs using a repeated-measures ANOVA. The assumptions for ANOVA, including normality and sphericity, were checked. The Shapiro-Wilk test confirmed normality ($p = .21$, and Mauchly's test for sphericity indicated no violations ($p = .17$). The two RAs, who were master's students specializing in mathematics

education, underwent rigorous training to ensure consistent and accurate classification of errors. The repeated-measures ANOVA results showed that there was no significant difference in error identification between the two RAs ($p > .05$), confirming the reliability of the chatbot designs' error patterns.

3.5 Result

Table 1 presents the error patterns identified by students using the Peer-Solution-Check and Step-by-Step chatbots, as well as those identified by the RAs. The table presents the average number of errors for each pattern, with corresponding standard deviations in parentheses.

The t -test results revealed a significant difference in the time students required to complete tasks between the two chatbot designs, $t(49) = -4.53$, $p = .014$. Specifically, students took significantly less time using the Peer-Solution-Check chatbot ($M = 15.43$, $SD = 1.20$) compared to the Guided-Problem-Solving chatbot ($M = 18.03$, $SD = 2.85$).

The repeated-measures ANOVA for error patterns revealed no significant differences in error pattern identification across the three methods (Worked, step-by-Step, and RAs). $F(2, 98) = 1.24$, $p = .305$.

Table 1: Error patterns identified by students using the Peer-Solution-Check and Step-by-Step chatbots, compared with the RA's identification.

Error Pattern	Peer-Solution-Check M (SD)	Step-by-Step M (SD)	Research Assistant M (SD)
General Comprehension	0.83 (0.21)	0.91 (0.32)	0.89 (0.01)
Mathematical Relationship Comprehension	0.32 (0.08)	0.27 (0.07)	0.31 (0.02)
Equation Construction	0.05 (0.10)	0.00 (0.12)	0.00 (0.02)
Calculation	0.01 (0.21)	0.02 (0.31)	0.08 (0.02)

Regarding students' perceived engagement and cognitive load between the *Peer-Solution-Check* and *Guided-Problem-Solving* conditions, the t-test indicated a significant difference in engagement, $t(49) = -3.67, p = .015$, with students reporting significantly lower engagement for *Guided-Problem-Solving* ($M = 4.11, SD = 1.24$) compared to *Peer-Solution-Check* ($M = 4.91, SD = 1.37$). A significant difference was also found in students' perceived cognitive load, $t(49) = 4.83, p < .001$, with *Guided-Problem-Solving* ($M = 3.22, SD = 1.33$) resulting in higher cognitive load compared to the *Peer-Solution-Check* ($M = 2.49, SD = 1.14$).

3.6 Discussions

The findings strongly indicate that the *Peer-Solution-Check* design is superior to the *Guided-Problem-Solving* design for error analysis, as it consistently outperforms the latter in terms of efficiency, engagement, and cognitive load, while maintaining equivalent reliability. These results suggest that, for the purpose of error analysis, the *Peer-Solution-Check* chatbot provides a more effective and user-friendly solution.

The *Peer-Solution-Check* design's advantage in efficiency was evident in the significantly shorter task completion times. By presenting fully solved problems for students to analyze, this approach streamlines the error-detection process, reducing the cognitive and temporal demands associated with generating responses. This aligns with cognitive load theory, which emphasizes the importance of minimizing extraneous cognitive demands to allow students to focus on core analytical tasks. The *Guided-Problem-Solving* design, in contrast, requires students to generate answers for every step, leading to longer task durations and higher cognitive strain. These findings underscore the practical benefits of the *Peer-Solution-Check* design in contexts where time and cognitive resources are constrained [13]. Engagement was another area where the *Peer-Solution-Check* design excelled. Students reported significantly higher engagement with this approach, which fosters a low-pressure environment for critical reflection and evaluation. Unlike the *Guided-Problem-Solving* design, which can feel rigid and evaluative due to its demand for constant responses, the *Peer-Solution-Check* design allows students to focus on understanding and analyzing errors without the stress of performance-based tasks. This dynamic likely contributed to the higher reported levels of engagement, highlighting the importance of reducing pressure to sustain student involvement and focus [16].

The *Peer-Solution-Check* chatbot also imposed a significantly lower cognitive load, further reinforcing its suitability for error analysis. By removing the need to generate responses, the design reduces the mental effort required to navigate the task, allowing students to allocate their cognitive resources toward understanding and error evaluation [17]. The *Guided-Problem-Solving* design, while encouraging active problem-solving, places greater cognitive demands on students as they must simultaneously identify errors, recall information, and formulate responses. This multitasking can overload cognitive resources, particularly in complex tasks, potentially detracting from the chatbot's effectiveness for error analysis.

Despite its strengths in engagement and cognitive efficiency, the *Guided-Problem-Solving* design remains reliable, as both approaches demonstrated no significant difference with RA's identification. This suggests that both designs are capable of accurately identifying error patterns, ensuring consistent feedback for educators and students. However, the *Guided-Problem-Solving* design's higher cognitive load and lower engagement may undermine its practical value, especially when used in time-sensitive or high-stress scenarios.

These findings underscore the clear advantage of the *Peer-Solution-Check* design for error analysis tasks. By offering a streamlined, engaging, and cognitively efficient approach, it meets the demands of error analysis more effectively than the *Guided-Problem-Solving* design. While the *Guided-Problem-Solving* chatbot's emphasis on active problem-solving may have value in specific contexts, such as skill-building or exploratory learning, its limitations in engagement and cognitive load make it less suitable for systematic error analysis. Future research could explore hybrid models that integrate the reflective benefits of *Peer-Solution-Check* with selective, targeted questioning to balance cognitive demands and foster deeper exploration. This study contributes to the growing body of research on AI-driven educational tools by demonstrating that aligning chatbot design with theoretical principles such as cognitive load theory can enhance their effectiveness. By focusing on reducing cognitive strain and promoting engagement, the *Peer-Solution-Check* design provides a robust framework for error analysis that meets the needs of both students and educators in diverse educational settings.

4 Limitations and Future Research

One limitation is that the study focused primarily on engagement, cognitive load, and task completion time within a controlled environment. Although the chatbots demonstrated high accuracy in error detection and notable efficiency in guiding students through tasks, future research should explore how these findings generalize to more dynamic, real-world educational settings where distractions, diverse learning objectives, and varying task complexities may influence performance. Additionally, studies comparing chatbot performance across different age groups and educational levels would help establish their broader applicability.

Another limitation is the small sample size of 50 fifth-grade students, which limits the generalizability of the findings. A larger and more diverse sample is necessary to apply these results to a wider student population. Additionally, the study did not explore how individual differences, such as students' prior knowledge and learning styles, might affect the effectiveness of the chatbot designs.

These factors could interact with the chatbot's role and questioning style, potentially influencing outcomes. Future research should investigate these individual differences to better understand how different users respond to various chatbot designs. Moreover, the study did not consider students' familiarity with technology, which could influence their interactions with the chatbot. Future research should investigate how varying levels of technological proficiency impact students' perceptions of chatbot designs and how these designs can be optimized for diverse user needs.

Lastly, this study focused on only two chatbot designs, limiting the exploration of other potentially effective approaches for detecting students' error patterns. Future research should investigate a wider variety of chatbot designs, incorporating different interaction styles and roles to gain a more complete understanding of how various designs affect student engagement, cognitive load, and error detection.

5 Conclusion

This study demonstrates the clear advantages of the Peer-Solution-Check chatbot design over the Guided-Problem-Solving approach for error analysis tasks. The Peer-Solution-Check design consistently outperformed its counterpart in efficiency, engagement, and cognitive load, while maintaining equivalent reliability in error detection. By reducing cognitive and temporal demands, the Peer-Solution-Check design provides a streamlined and user-friendly framework that fosters critical reflection without the stress of constant performance-based tasks. In contrast, the Guided-Problem-Solving design, while reliable and promoting active problem-solving, imposes greater cognitive strain and lower engagement, limiting its practical value in high-demand scenarios.

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A Word problems in the two chatbots

- problem 1: Xiao Li bought some pears and grapes at the market. The pears cost 2.5 yuan, and the grapes cost 3.6 yuan. How much did he spend in total?
- problem 2: Uncle Zhang's garden yielded a large amount of oranges today. He gave 10 oranges to Xiao Mei and 8 oranges to Xiao Jun. How many oranges does he have left?
- problem 3: Xiao Hong went to the bookstore and bought a notebook for 12.5 yuan and a pen for 3.3 yuan. How much did she spend in total?
- problem 4: In the market, a watermelon costs 5.5 yuan, and a pineapple costs 7.8 yuan. If Xiao Zhao bought one watermelon and two pineapples, how much did he spend in total?
- problem 5: At the fruit stand, the strawberries cost 6.3 yuan per kilogram, and the blueberries cost 8.2 yuan per kilogram. If Xiao Yuan bought 2 kilograms of strawberries and 1 kilogram of blueberries, how much did he spend in total?

B The Self-report Measure of Engagement and Cognitive Load

	Item	Strongly Disagree	Disagree	Somewhat Disagree	Neither Agree nor Disagree	Somewhat Agree	Agree	Strongly Agree
Step-by-Step Design								
1	I like talking with Miss Wang.							
2	Talking with Miss Wang feels the most interesting.							
3	I am not willing to talk with Miss Wang.							
4	I feel curious about the conversation with Miss Wang.							
5	What Miss Wang says makes me more willing to respond.							
6	I will carefully read what Miss Wang says on.							
7	I will actively think about the questions Miss Wang asks .							
8	I will try my best to respond to the requests Miss Wang makes .							
9	I do not have the patience to read what Miss Wang says .							
10	I do not want to answer the questions Miss Wang asks on .							
11	I need to stay fully focused to answer Miss Wang's questions .							
12	While talking with Miss Wang, I have to think about many things.							
13	The conversation with Miss Wang leaves me exhausted.							
14	I need to read Miss Wang's messages on multiple times before I can respond.							
15	What Miss Wang says on feels too long for me.							
Peer-Solution-Check Design								
1	I like talking with Zihan.							
2	Talking with Zihan feels the most interesting.							
3	I am not willing to talk with Zihan .							
4	I feel curious about the conversation with Zihan .							
5	What Zihan says makes me more willing to respond.							
6	I will carefully read what Zihan says .							
7	I will actively think about the questions Zihan asks.							
8	I will try my best to respond to the requests Zihan makes .							
9	I do not have the patience to read what Zihan says on .							
10	I do not want to answer the questions Zihan asks on .							
11	I need to stay fully focused to answer Zihan's questions .							
12	While talking with Zihan , I have to think about many things.							
13	The conversation with Zihan leaves me exhausted.							
14	I need to read Zihan's messages multiple times before I can respond.							
15	What Zihan says feels too long for me.							

Note. In the measures related to each chatbot, items 1-10 assess engagement, and items 11-15 assess cognitive load.